

Introduction to Computer Vision

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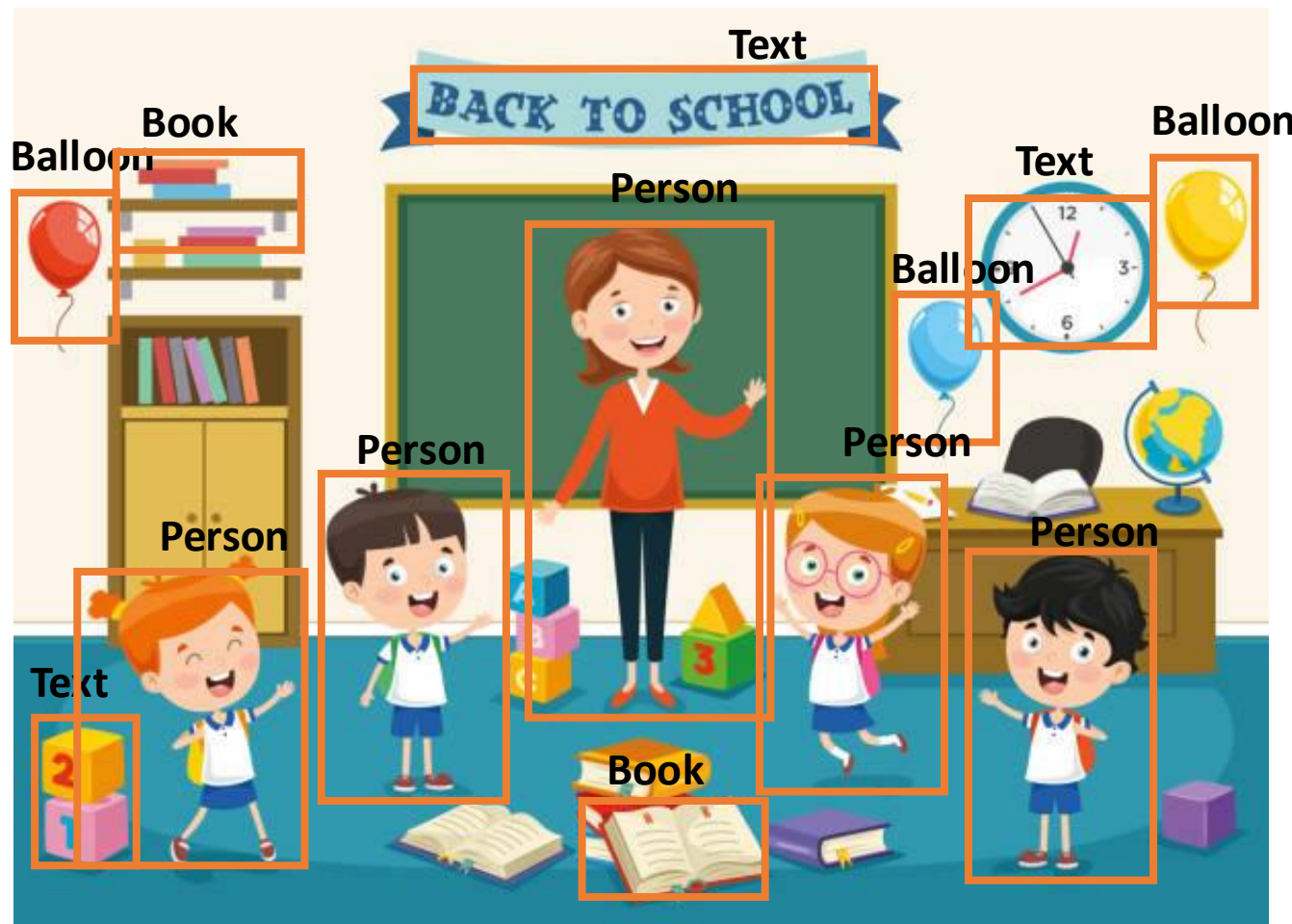
Lecture 1

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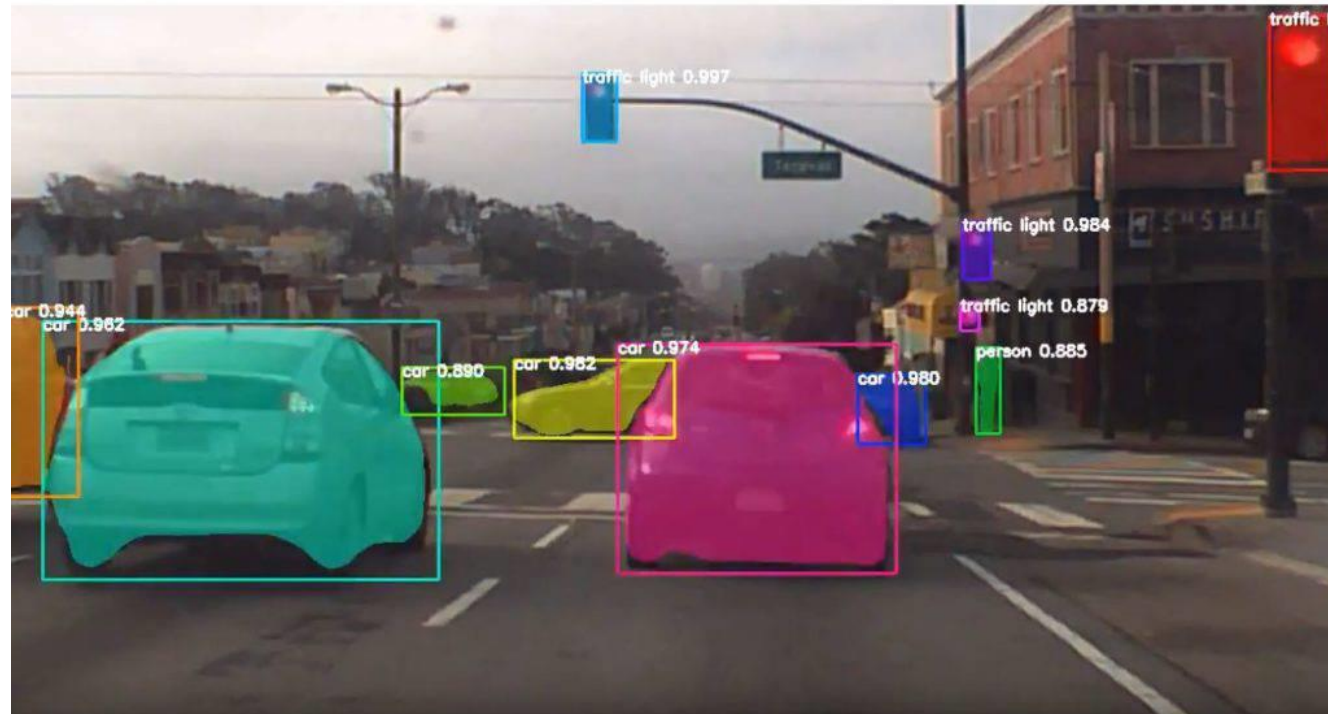
Computer Vision

- What is computer vision?
 - The computer should be able to answer all questions related to images similar to humans.



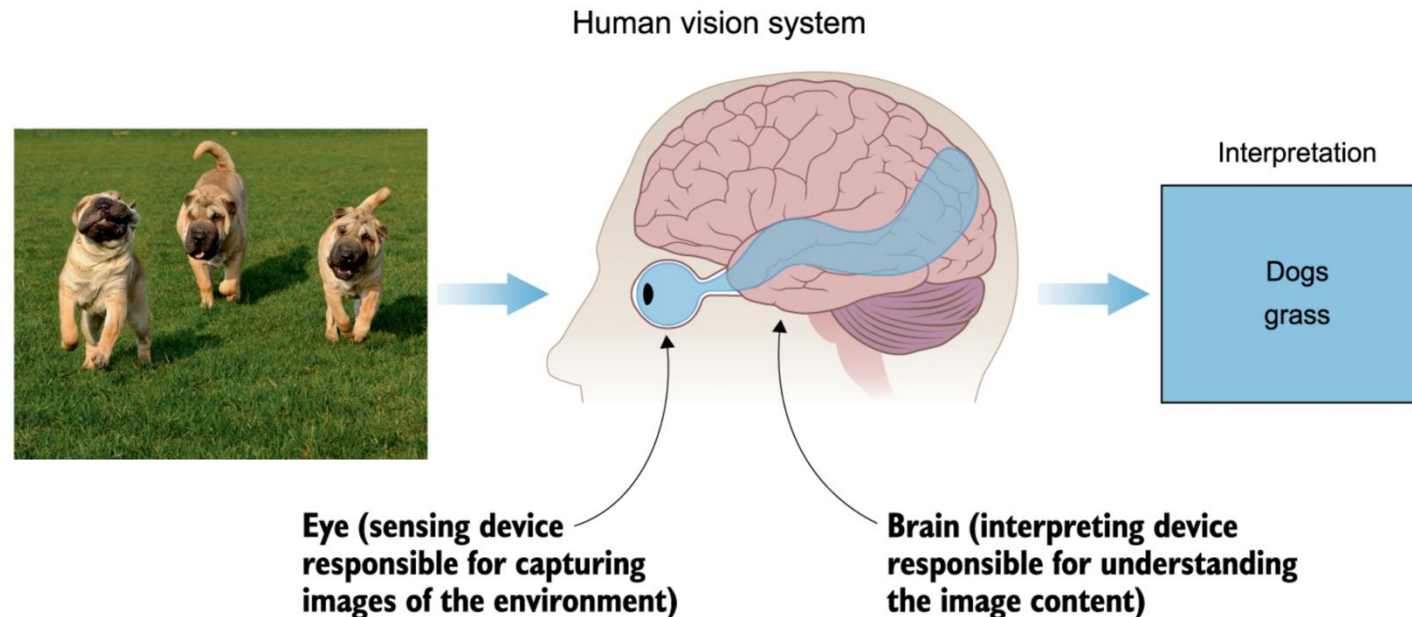
Computer Vision

- Visual Perception
 - Observing and analyzing the objects and the related patterns in the images or videos
 - Understand the environment through visual input
 - E.g. cameras in autonomous vehicles



Vision Systems

- Image processing
 - Difference between image processing and understanding what's happening in the image
 - Part of the complex system to interpret the image content
- Human Vision Systems



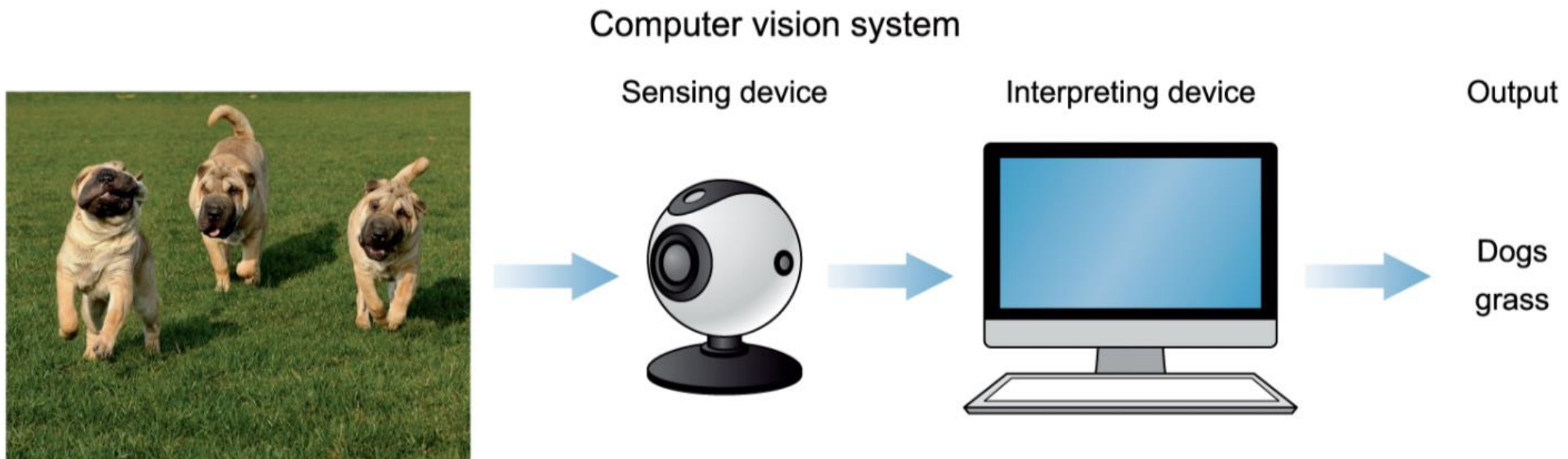
Human Vision System – ML Perspective

- Supervised learning - Brain learns to identify dogs in the images by looking at examples of several dog-labeled images.
- Labeled data – data for which you already know the answer. E.g. images of the dog
- If you see a different object e.g. horse, your brain will classify that as dog as it has never seen horse image before. When the brain is told that its prediction is wrong, it will adjust the parameters to learn the features of the horse. It will keep on tuning the horse features until it predicts it correctly.

Hurray! Your brain is now trained to recognize horse as well!

Vision Systems

- AI Vision Systems





Sensing devices

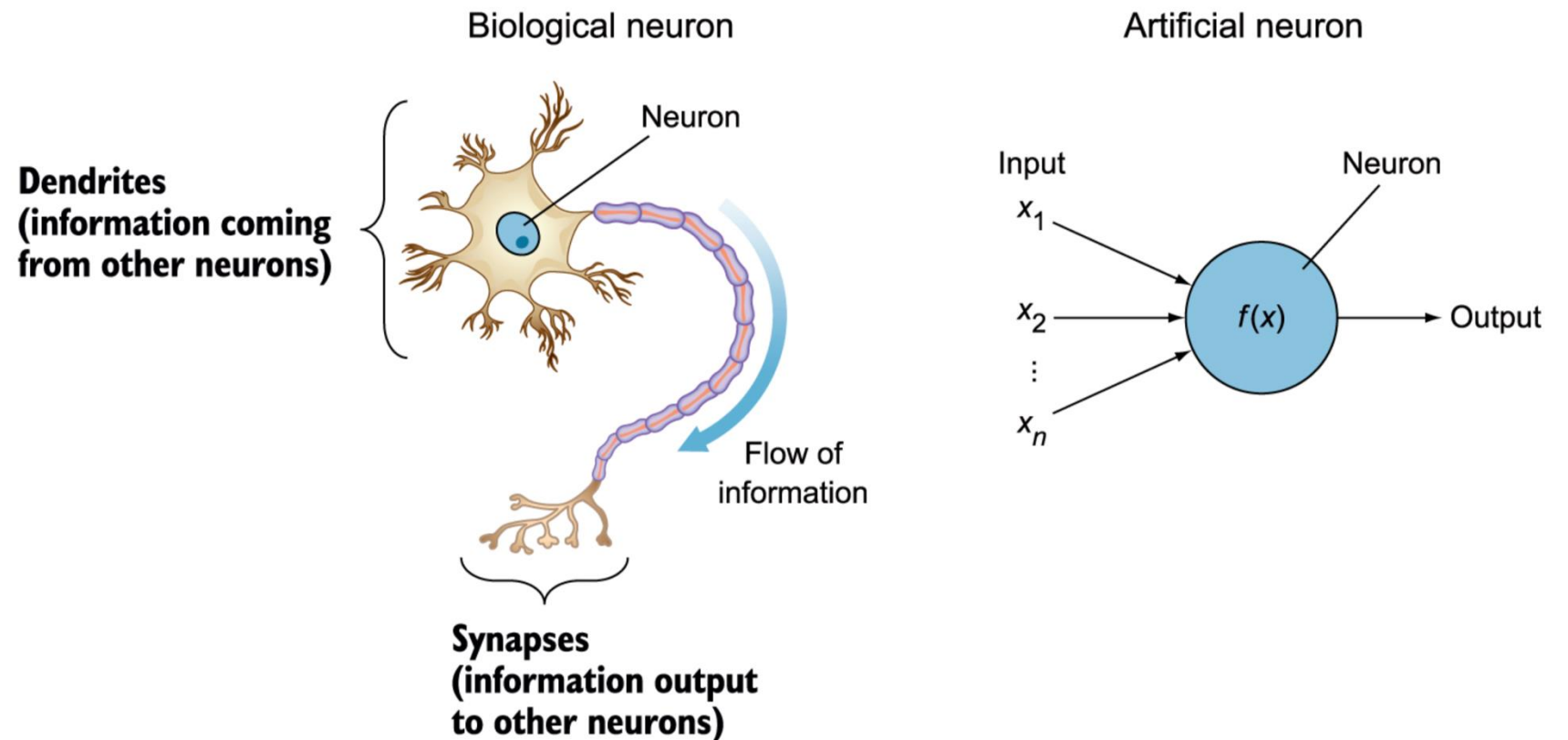
- Camera
- Radar – measures distance and velocity but cannot see the finer details
- Lidar – Light detection and ranging - uses ultraviolet infrared light pulses to create high resolution 3D map of the surrounding area
- X-ray and CT scan for medical diagnosis applications
- Vision sensors in combination with non-vision sensors for understanding the environment



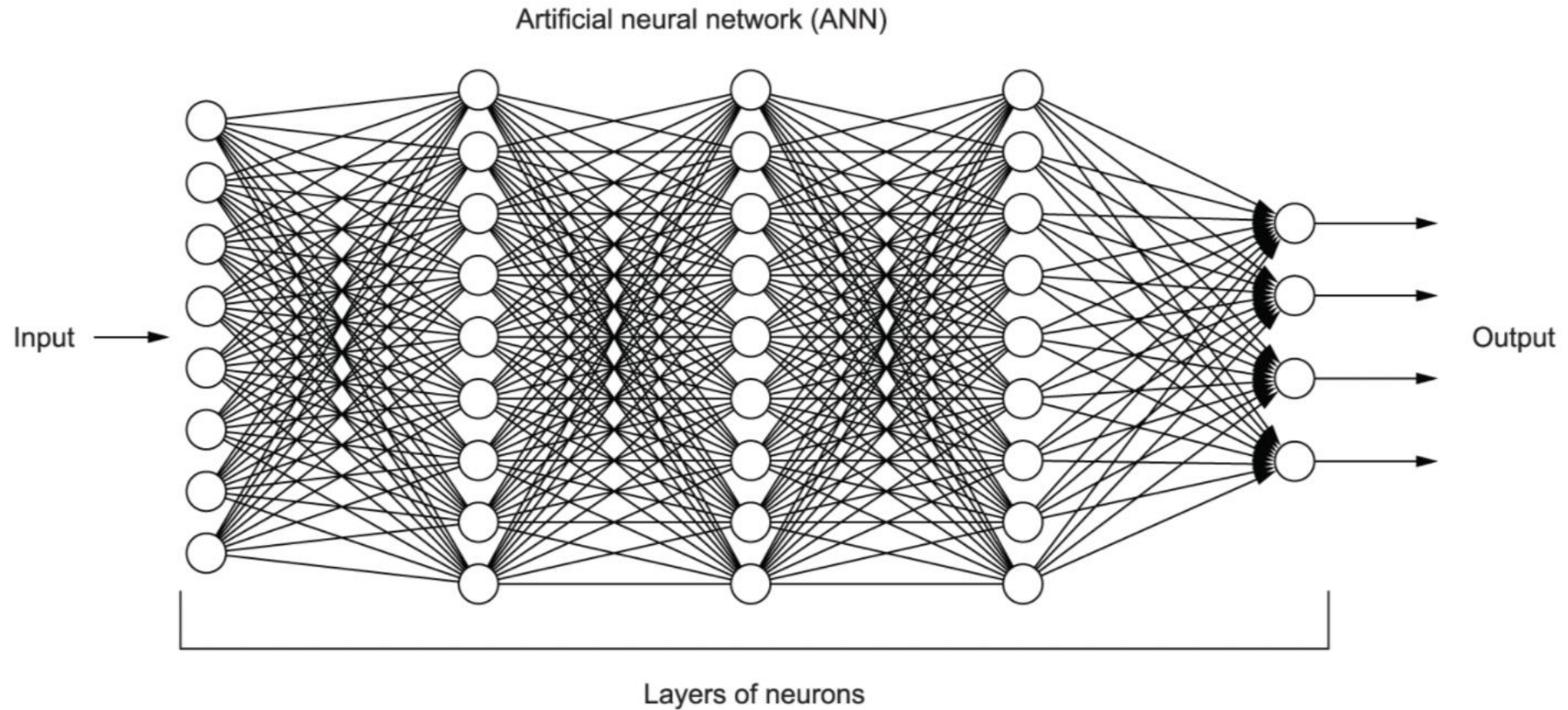
Interpreting devices

- Computer Vision Algorithms
 - Take image from the sensing device and learn features and patterns to identify the objects

Fig. Similarities between biological neurons and artificial systems



Stacked Layers on Neurons



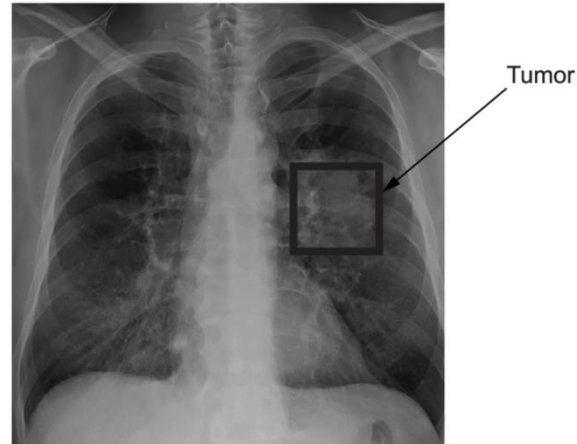
Deep Learning Model

Applications of Computer Vision

Image classification



CT scan



X-ray

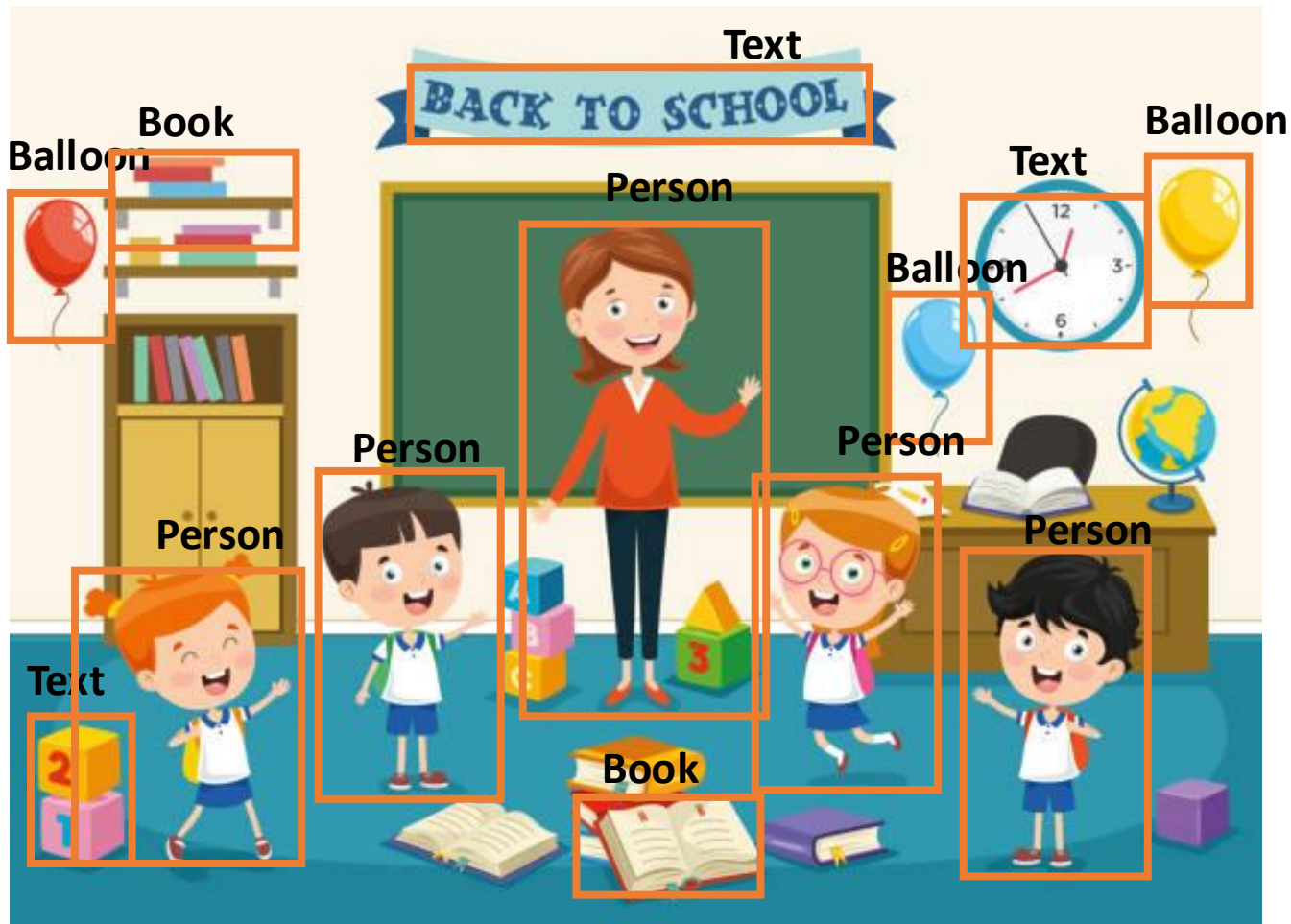
Lung cancer detection



Traffic sign detection

Applications of Computer Vision

Object detection and localization



Applications of Computer Vision

Generating art (style transfer)

Original image



+

Style



=

Generated art



Applications of Computer Vision

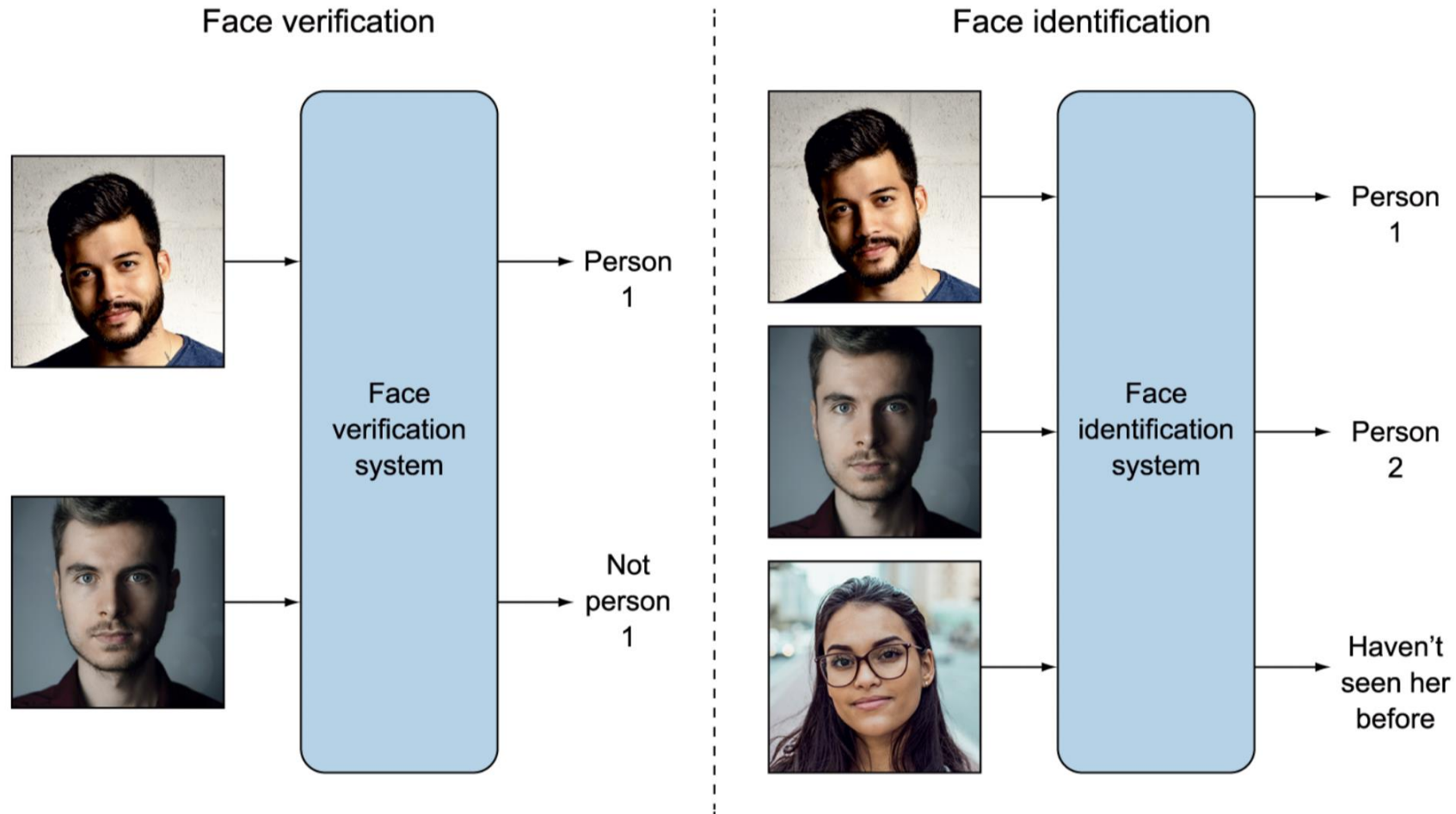
Generative adversarial networks (GANs)

- can create new, “made-up” images from a set of existing images.
- E.g. fake human faces [1]



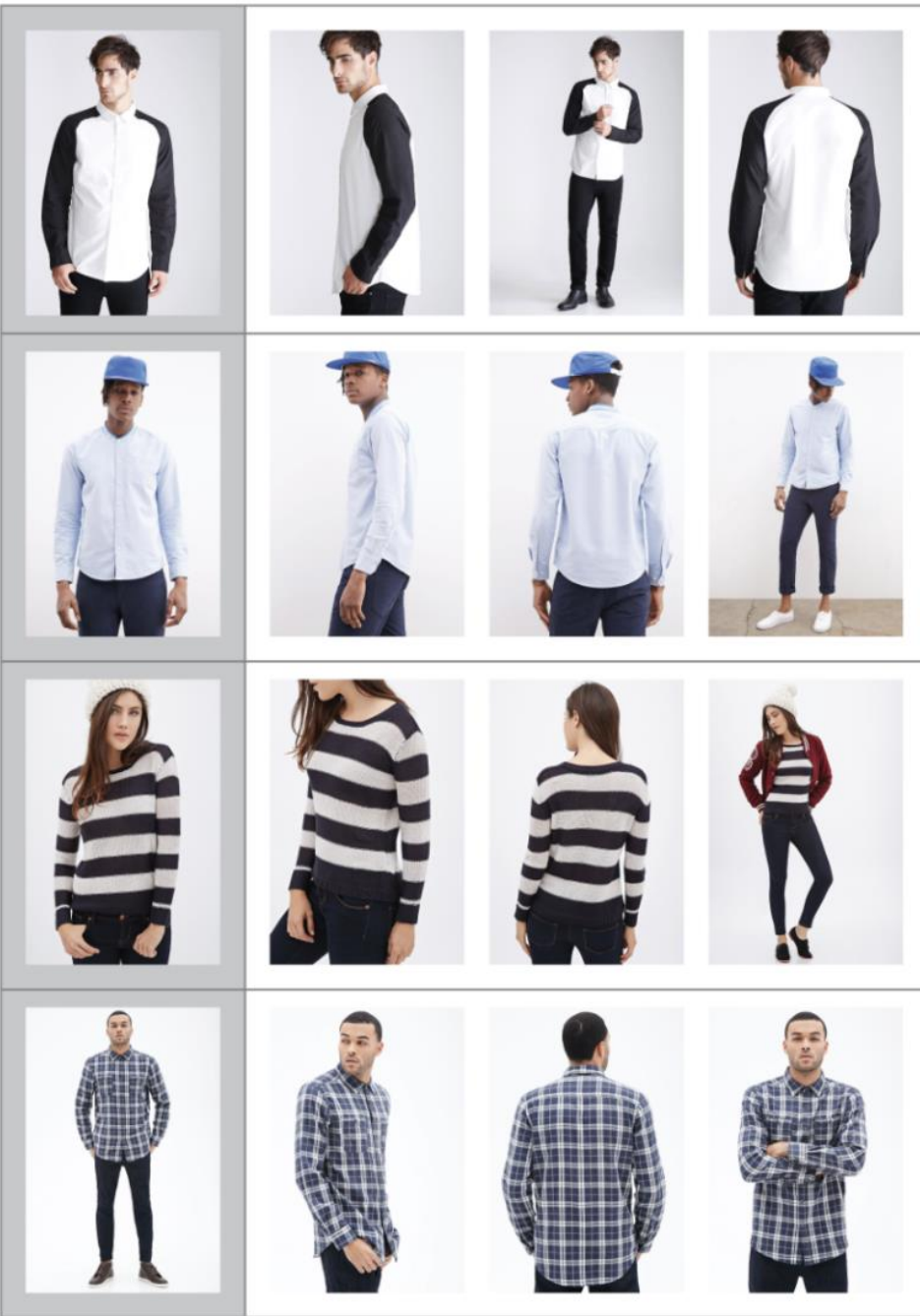
Applications of Computer Vision

Face Recognition



Query

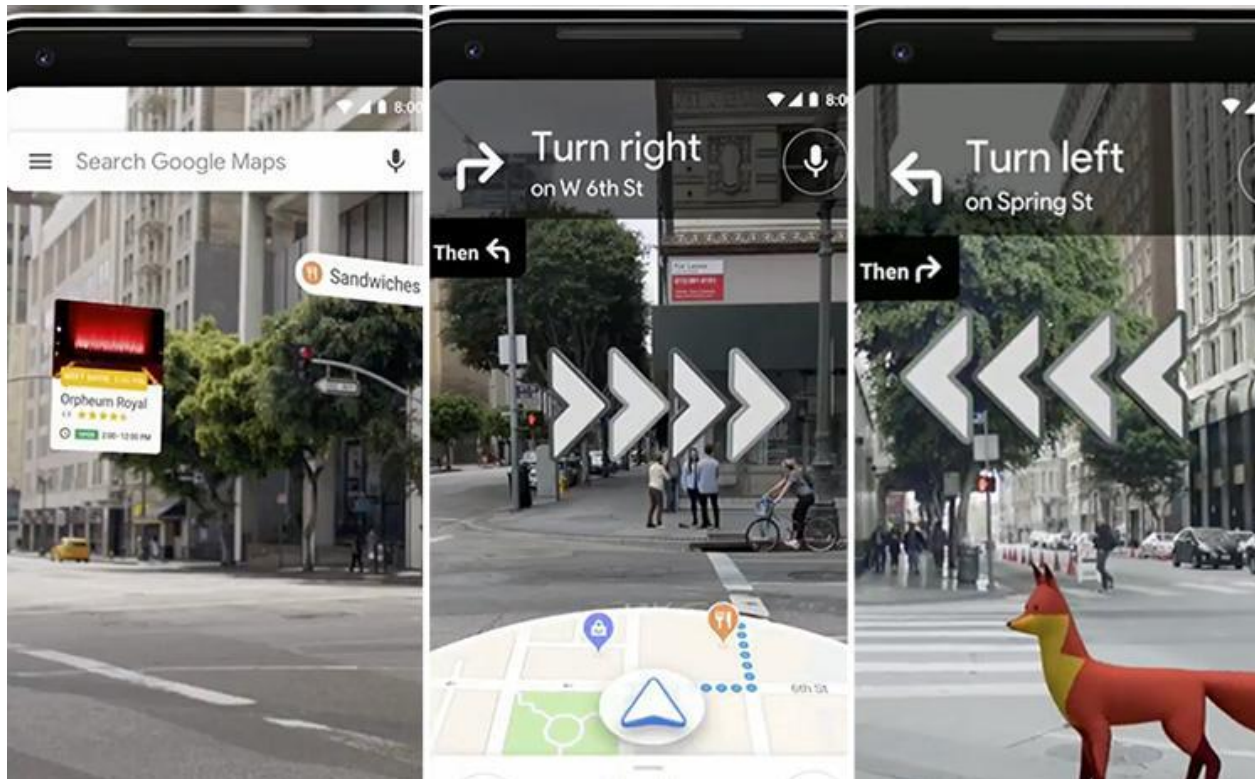
Retrievals



Applications of Computer Vision

- Image recommendation system

Visual Positioning System (VPS)



- Accurate location prediction
 - Satellite positioning (GPS) + Landmarks + Street signs etc
- Walking/direction buddy
 - AR based buddy

Reference

- Book – Deep Learning for Vision Systems by Mohamad Elgendy, Manning Publication, November 2020
- <https://www.carscoops.com/2023/01/mercedes-becomes-first-automaker-to-get-approval-for-level-3-autonomy-in-the-u-s/>